

Group Members:

1. Ali Asghar (2059)
2. Shahad Bangash (1980)
3. Suleman Shah (1983)

SHOW TABLES Command:

show tables command display the list of all the tables in our databases as shown in figure 1-1

```
mysql> show tables;
+-----+
| Tables_in_hallbookingsystem |
+-----+
| conference_center            |
| customer                    |
| hall                         |
| invoice                     |
| manager                     |
| manages                      |
| owner                       |
| review                      |
| wedding_hall                |
+-----+
9 rows in set (0.01 sec)
```

Figure 1-1, All tables in Hall Booking System Database

DESCRIBE TABLE Commands:

Following are the describe commands for all the tables present in the database.

```
mysql> describe customer;
```

Field	Type	Null	Key	Default	Extra
customer_id	int	NO	PRI	NULL	
name	varchar(50)	NO		NULL	
no_of_halls_booked	int	YES		0	
Phone_No	int	YES		NULL	
address	varchar(30)	YES		NULL	

Figure 2-1, Metadata of customer table in Hall Booking System Database

```
mysql> describe hall;
```

Field	Type	Null	Key	Default	Extra
hall_id	int	NO	PRI	NULL	
name	varchar(50)	NO		NULL	
Location	varchar(50)	NO		NULL	
cust_id	int	YES	MUL	0	
Booking_Price	int	YES		NULL	
Capacity	int	YES		NULL	
is_Available	tinyint(1)	YES		NULL	
booking_start_time	date	YES		NULL	
booking_end_time	date	YES		NULL	
hall_type	varchar(50)	YES		NULL	
owner_id	int	YES	MUL	NULL	

11 rows in set (0.01 sec)

Figure 2-2, Metadata of hall table in Hall Booking System Database

```
mysql> describe invoice;
```

Field	Type	Null	Key	Default	Extra
invoice_no	int	NO	PRI	NULL	
name	varchar(50)	NO		NULL	
cus_id	int	YES	MUL	NULL	
PaymentPaid	int	YES		NULL	
date	date	YES		NULL	

```
5 rows in set (0.01 sec)
```

Figure 2-3, Metadata of invoice table in Hall Booking System Database

```
mysql> describe manager;
```

Field	Type	Null	Key	Default	Extra
manager_id	int	NO	PRI	NULL	
name	varchar(50)	NO		NULL	
no_of_halls_booked	int	YES		0	
Phone_No	int	YES		NULL	

```
4 rows in set (0.01 sec)
```

Figure 2-4, Metadata of manager table in Hall Booking System Database

```
mysql> describe manages;
```

Field	Type	Null	Key	Default	Extra
manager_id	int	NO	PRI	NULL	
hall_id	int	NO	PRI	NULL	

```
2 rows in set (0.01 sec)
```

Figure 2-5, Metadata of manages table in Hall Booking System Database

```
mysql> describe owner;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| owner_id   | int           | NO   | PRI | NULL    |       |
| name       | varchar(50)   | NO   |     | NULL    |       |
| phone      | int           | YES  |     | NULL    |       |
| address    | varchar(50)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.01 sec)
```

Figure 2-6, Metadata of owner table in Hall Booking System Database

```
mysql> describe review;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| review_id  | int           | NO   | PRI | NULL    |       |
| title      | varchar(50)   | NO   |     | NULL    |       |
| Body       | varchar(50)   | NO   |     | NULL    |       |
| customer_id | int           | YES  | MUL | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.01 sec)
```

Figure 2-7, Metadata of review table in Hall Booking System Database

```
mysql> describe wedding_hall;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| hall_id    | int           | NO   | PRI | NULL    |       |
| Chef       | varchar(50)   | YES  |     | NULL    |       |
| PricePerHour | int           | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.01 sec)
```

Figure 2-8, Metadata of wedding_hall table in Hall Booking System Database

```
mysql> describe conference_center;
+-----+-----+-----+-----+-----+-----+
| Field          | Type  | Null  | Key  | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| hall_id        | int   | NO    | PRI  | NULL    |       |
| PricePerDay    | int   | YES   |      | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.01 sec)
```

Figure 2-9, Metadata of conference_center table in Hall Booking System Database

SHOW CREATE TABLE Command:

Following are the create table commands for all the tables present in the database.

```
| customer | CREATE TABLE `customer` (
  `customer_id` int NOT NULL,
  `name` varchar(50) NOT NULL,
  `no_of_halls_booked` int DEFAULT '0',
  `Phone_No` int DEFAULT NULL,
  `address` varchar(30) DEFAULT NULL,
  PRIMARY KEY (`customer_id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.02 sec)
```

Figure 3-1, create table command for customer table in Hall Booking System Database

```

| hall | CREATE TABLE `hall` (
  `hall_id` int NOT NULL,
  `name` varchar(50) NOT NULL,
  `Location` varchar(50) NOT NULL,
  `cust_id` int DEFAULT '0',
  `Booking_Price` int DEFAULT NULL,
  `Capacity` int DEFAULT NULL,
  `is_Available` tinyint(1) DEFAULT NULL,
  `booking_start_time` date DEFAULT NULL,
  `booking_end_time` date DEFAULT NULL,
  `hall_type` varchar(50) DEFAULT NULL,
  `owner_id` int DEFAULT NULL,
  PRIMARY KEY (`hall_id`),
  KEY `owner_id` (`owner_id`),
  KEY `cust_id` (`cust_id`),
  CONSTRAINT `hall_ibfk_1` FOREIGN KEY (`owner_id`) REFERENCES `owner` (`owner_id`),
  CONSTRAINT `hall_ibfk_2` FOREIGN KEY (`cust_id`) REFERENCES `customer` (`customer_id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+

```

Figure 3-2, create table command for hall table in Hall Booking System Database

```

| invoice | CREATE TABLE `invoice` (
  `invoice_no` int NOT NULL,
  `name` varchar(50) NOT NULL,
  `cus_id` int DEFAULT NULL,
  `PaymentPaid` int DEFAULT NULL,
  `date` date DEFAULT NULL,
  PRIMARY KEY (`invoice_no`),
  KEY `cus_id` (`cus_id`),
  CONSTRAINT `invoice_ibfk_1` FOREIGN KEY (`cus_id`) REFERENCES `customer` (`customer_id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+

```

Figure 3-3, create table command for invoice table in Hall Booking System Database

```

| manager | CREATE TABLE `manager` (
  `manager_id` int NOT NULL,
  `name` varchar(50) NOT NULL,
  `no_of_halls_booked` int DEFAULT '0',
  `Phone_No` int DEFAULT NULL,
  PRIMARY KEY (`manager_id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+

```

Figure 3-4, create table command for manager table in Hall Booking System Database

```

manages | CREATE TABLE `manages` (
  `manager_id` int NOT NULL,
  `hall_id` int NOT NULL,
  PRIMARY KEY (`manager_id`,`hall_id`),
  KEY `hall_id` (`hall_id`),
  CONSTRAINT `manages_ibfk_1` FOREIGN KEY (`manager_id`) REFERENCES `manager` (`manager_id`),
  CONSTRAINT `manages_ibfk_2` FOREIGN KEY (`hall_id`) REFERENCES `hall` (`hall_id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |

```

Figure 3-5, create table command for manages table in Hall Booking System Database

```

owner | CREATE TABLE `owner` (
  `owner_id` int NOT NULL,
  `name` varchar(50) NOT NULL,
  `phone` int DEFAULT NULL,
  `address` varchar(50) DEFAULT NULL,
  PRIMARY KEY (`owner_id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |

```

Figure 3-6, create table command for owner table in Hall Booking System Database

```

review | CREATE TABLE `review` (
  `review_id` int NOT NULL,
  `title` varchar(50) NOT NULL,
  `Body` varchar(50) NOT NULL,
  `customer_id` int DEFAULT NULL,
  PRIMARY KEY (`review_id`),
  KEY `customer_id` (`customer_id`),
  CONSTRAINT `review_ibfk_1` FOREIGN KEY (`customer_id`) REFERENCES `customer` (`customer_id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |

```

Figure 3-7, create table command for review table in Hall Booking System Database

```

wedding_hall | CREATE TABLE `wedding_hall` (
  `hall_id` int NOT NULL,
  `Chef` varchar(50) DEFAULT NULL,
  `PricePerHour` int DEFAULT NULL,
  PRIMARY KEY (`hall_id`),
  CONSTRAINT `wedding_hall_ibfk_1` FOREIGN KEY (`hall_id`) REFERENCES `hall` (`hall_id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |

```

Figure 3-8, create table command for wedding_hall table in Hall Booking System Database

```

conference_center | CREATE TABLE `conference_center` (
  `hall_id` int NOT NULL,
  `PricePerDay` int DEFAULT NULL,
  PRIMARY KEY (`hall_id`),
  CONSTRAINT `conference_center_ibfk_1` FOREIGN KEY (`hall_id`) REFERENCES `hall` (`hall_id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |

```

Figure 3-9, create table command for conference_center table in Hall Booking System Database

SELECT * FROM TABLE Command:

Following are the select * from table commands for all the tables present in the database.

```
mysql> select * from customer;
+-----+-----+-----+-----+-----+
| customer_id | name       | no_of_halls_booked | Phone_No | address          |
+-----+-----+-----+-----+-----+
| 1 | Ali Ahmad | 2 | 1234567890 | Rahatabad St 10 |
| 2 | Adnan Khan | 1 | 2147483647 | Phase 5 Hayatabad |
| 3 | Ismail Khan | 0 | 2147483647 | Pishtakhara Chowk |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

Figure 4-1 select * command for customer table in Hall Booking System Database

```
mysql> select * from hall;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| hall_id | name       | Location | cust_id | Booking_Price | Capacity | is_Available | booking_start_time | booking_end_time | hall_type | owner_id |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | Grand Ballroom | Downtown | 1 | 5000 | 200 | 1 | 2024-06-01 | 2024-06-02 | Wedding | 1 |
| 2 | Conference Hall | Uptown | 2 | 3000 | 150 | 1 | 2024-06-10 | 2024-06-11 | Conference | 2 |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql>
```

Figure 4-2 select * command for hall table in Hall Booking System Database

```
mysql> select * from invoice;
+-----+-----+-----+-----+-----+
| invoice_no | name       | cus_id | PaymentPaid | date       |
+-----+-----+-----+-----+-----+
| 1 | Yar Afzal | 1 | 5000 | 2024-06-01 |
| 2 | Ismail | 2 | 3000 | 2024-06-10 |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

Figure 4-3 select * command for invoice table in Hall Booking System Database


```
mysql> select * from manager;
```

manager_id	name	no_of_halls_booked	Phone_No
1	Rehmat Ullah	1	2147483647
2	Farooq Ali	1	2147483647

```
2 rows in set (0.00 sec)
```

Figure 4-4 select * command for manager table in Hall Booking System Database

```
mysql> select * from manages;
```

manager_id	hall_id
1	1
2	2

```
2 rows in set (0.00 sec)

mysql> _
```

Figure 4-5 select * command for manages table in Hall Booking System Database

```
mysql> select * from owner;
```

owner_id	name	phone	address
1	Suleman Shah	2147483647	Ring Road Pishtakhara Chowk
2	Shayan Khan	2147483647	Phase 5 Hayatabad

```
2 rows in set (0.00 sec)
```

Figure 4-6 select * command for owner table in Hall Booking System Database

```
mysql> select * from review;
```

review_id	title	Body	customer_id
1	Great Experience	The hall was excellent and the service was great.	1
2	Good Service	The booking process was smooth and hassle-free.	2
3	Average Experience	The hall was okay but could be better.	3

```
3 rows in set (0.00 sec)
```

Figure 4-7 select * command for review table in Hall Booking System Database

```
mysql> select * from wedding_hall;
+-----+-----+-----+
| hall_id | Chef      | PricePerHour |
+-----+-----+-----+
|        1 | Chef Jan  |          100 |
+-----+-----+-----+
1 row in set (0.00 sec)
```

Figure 4-8 select * command for wedding_hall table in Hall Booking System Database

```
mysql> select * from conference_center;
+-----+-----+
| hall_id | PricePerDay |
+-----+-----+
|        2 |          500 |
+-----+-----+
1 row in set (0.00 sec)
```

Figure 4-9 select * command for conference_center table in Hall Booking System Database

SELECT * FROM TABLE Command with Variations:

Retrieve All Customers and Their Bookings

```
mysql> SELECT Customer.name AS customer_name, Hall.name AS hall_name, Hall.booking_start_time, Hall.booking_end_time FROM Customer JOIN Hall ON Customer.customer_id = Hall.cust_id;
+-----+-----+-----+-----+
| customer_name | hall_name      | booking_start_time | booking_end_time |
+-----+-----+-----+-----+
| Ali Ahmad     | Grand Ballroom | 2024-06-01         | 2024-06-02       |
| Adnan Khan    | Conference Hall | 2024-06-10         | 2024-06-11       |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

Figure 5-1, Retrieve All Customers and Their Bookings in Hall Booking System Database

Find All Halls Managed by a Specific Manager

```
mysql> SELECT Manager.name AS manager_name, Hall.name AS hall_name, Hall.Location FROM Manager JOIN Manages ON Manager.manager_id = Manages.manager_id JOIN Hall ON Manages.hall_id = Hall.hall_id WHERE Manager.name = 'Rehmat Ullah';
+-----+-----+-----+
| manager_name | hall_name      | Location |
+-----+-----+-----+
| Rehmat Ullah | Grand Ballroom | Downtown |
+-----+-----+-----+
1 row in set (0.00 sec)
```

Figure 5-2, Find All Halls Managed by a Specific Manager in Hall Booking System Database

List all reviews submitted by Customer

```
mysql> SELECT Customer.name AS customer_name, Review.title, Review.Body FROM Review JOIN Customer ON Review.customer_id = Customer.customer_id;
```

customer_name	title	Body
Ali Ahmad	Great Experience	The hall was excellent and the service was great.
Adnan Khan	Good Service	The booking process was smooth and hassle-free.
Ismail Khan	Average Experience	The hall was okay but could be better.

```
3 rows in set (0.00 sec)
```

Figure 5-3, List All reviews submitted by Customer in Hall Booking System Database

Retrieve All Invoices for a Specific Customer

```
mysql> SELECT Customer.name AS customer_name, Invoice.invoice_no, Invoice.PaymentPaid, Invoice.date FROM Invoice JOIN Customer ON Invoice.cus_id = Customer.customer_id WHERE Customer.name = 'Ali Ahmad';
```

customer_name	invoice_no	PaymentPaid	date
Ali Ahmad	1	5000	2024-06-01

```
1 row in set (0.00 sec)
```

```
mysql>
```

Figure 5-4, Retrieve All Invoices for a Specific Customer in Hall Booking System Database

Retrieve List of all available halls

```
mysql> SELECT
->     name AS hall_name,
->     Location,
->     Booking_Price,
->     Capacity
-> FROM
->     Hall
-> WHERE
->     is_Available = TRUE;
```

hall_name	Location	Booking_Price	Capacity
Grand Ballroom	Downtown	5000	200
Conference Hall	Uptown	3000	150

```
2 rows in set (0.00 sec)
```

```
mysql> _
```

Figure 5-5, Retrieve List of all available halls in Hall Booking System Database